ScholarPath — The Conceptual ERD (Step 1)

A Four-Part Data-Modelling Series: Part 1 of 4

ScholarPath Team

Abstract

This is the first instalment of a four-part teaching series that walks the ScholarPath data model through the classic database-design pipeline: **plain ERD → Enhanced ER (EERD) → Relational Mapping → Class Diagrams**. The present document establishes the *conceptual* starting point — a deliberately plain Entity–Relationship Diagram (ERD) drawn in Chen / Elmasri notation. We introduce the ScholarPath platform and its four actors, recall the vocabulary of the ER model (entities, relationships, attributes, keys and the $1/N/M$ cardinality ratios), and then present the basic conceptual diagram of the core entities and their relationships. Crucially, this is the "before" picture: at this stage the model is intentionally flat — a single `USER` entity with role as a mere attribute, no specialization, no weak entities, and no multivalued or derived attributes. Those simplifications are exactly what *Part 2: The Enhanced ER Diagram* sets out to repair, so this document closes by motivating that enhancement.

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1 Introduction to the ScholarPath System

ScholarPath is a web platform that connects students seeking educational funding with the people and organisations that can help them obtain it. In a single application it brings together a scholarship catalogue and application tracker, a consultant-booking and session-recording service, a scholarship-provider review service for applications, a community forum with private chat, a resources hub, and a layer of analytics and AI-assisted recommendation. The platform runs in *free mode* — all services are offered at no charge, so there is no money-settlement subsystem in the model. The system is built as a .NET backend with a React front end; this series, however, is concerned only with the *data model* that underpins it.

1.1 The four actors

Every interaction in ScholarPath is performed by one of four kinds of user. In the running system these are roles granted to a single account principal, but conceptually they behave as four distinct actors:

- **Student** — the primary beneficiary. A student maintains an academic profile (academic level, field of study, GPA, preferred countries), discovers and applies to scholarships, uploads supporting documents, books consultant sessions, requests scholarship-provider reviews of an application, and participates in the community.
- **ScholarshipProvider** — an organisation that owns and publishes scholarships, and that can review a student's application. A provider carries an organisation legal name, a verification status and an aggregate rating.
- **Consultant** — a domain expert who publishes availability, conducts one-to-one booking sessions, and is rated afterwards. A consultant carries a session duration, expertise tags and a professional title.
- **Admin** — the platform operator who moderates content, approves verification and upgrade requests, and configures the platform.

1.2 Main subject areas

The platform's data naturally clusters into six subject areas, all radiating from the user. These same clusters reappear, in much greater detail, in the later parts of this series:

1. **Identity & Profile** — accounts, roles and profile data.
2. **Scholarships, Applications & Documents** — the catalogue, the applications students submit against it, the documents that support them, and scholarship-provider reviews.
3. **Consultant Booking & Ratings** — consultant availability, bookings, and the reviews that follow.
4. **Community & Chat** — forum posts and private conversations.

5. **Resources & Notifications** — learning resources and the notifications that keep users informed.
6. **AI, Knowledge & Platform** — recommendation, the knowledge base, auditing and configuration.

This first document zooms out to the most important entities across these areas and draws them plainly. The companion document *Part 2: The Enhanced ER Diagram* adds the structural refinements; *Part 3: Relational Mapping* turns the model into tables; and *Part 4: Class Diagrams* expresses it in object-oriented terms.

2 What an ERD Is

The **Entity–Relationship Diagram (ERD)** is the standard conceptual tool for describing the structure of data independently of any particular database technology. It was introduced by Peter Chen and is presented here in the **Chen / Elmasri** notation used throughout most database textbooks. The notation rests on four building blocks.

2.1 Entities and entity types

An **entity** is a real-world object or concept about which we store data — a particular scholarship, a particular booking, a particular user. An **entity type** is the named category to which such objects belong (for example `SCHOLARSHIP`). In Chen notation an entity type is drawn as a **rectangle**.

2.2 Attributes

An **attribute** is a property of an entity type — a scholarship has a title, a deadline and a status. Attributes are drawn as **ovals** connected to their entity rectangle. The conceptual model deliberately records only the few attributes that matter for understanding the structure, not the full column list (that is the job of Part 3).

2.3 Keys

A **key** attribute uniquely identifies each entity of a type — no two entities may share the same key value. In Chen notation a key attribute is **underlined**. In ScholarPath every core entity is keyed by an `Id`, shown underlined in the diagram.

2.4 Relationships

A **relationship** is a meaningful association between entities — a user *submits* an application; a booking is *rated by* a consultant review. A **relationship type** is drawn as a **diamond** on the line that joins the participating entity rectangles, labelled with a verb.

2.5 Cardinality ratios: 1, N and M

The **cardinality ratio** of a binary relationship records how many entities of one type may be associated with the other. Each end of the relationship line is marked with a symbol:

- **1** — "one". At most one entity participates on this side.

- **N** (or **M**) — "many". Any number of entities may participate on this side.

This yields the three classic shapes summarised in Table 1.

Table 1: The three cardinality ratios in Chen notation.

Ratio	Reads as	ScholarPath example
1 : 1 (one-to-one)	one A relates to one B	a booking is rated by at most one consultant review
1 : N (one-to-many)	one A relates to many B	one user submits many applications
M : N (many-to-many)	many A relate to many B	a user participates in many conversations, each with many

3 The Basic Conceptual ERD

Figure 1 presents the basic conceptual ERD for ScholarPath in authentic Chen notation: rectangles are entity types, diamonds are relationship types, ovals are attributes, underlined attributes are keys, and the `1`/`N`/`M` markers on the edges give the cardinality ratios. It is the source diagram for everything that follows in this series.

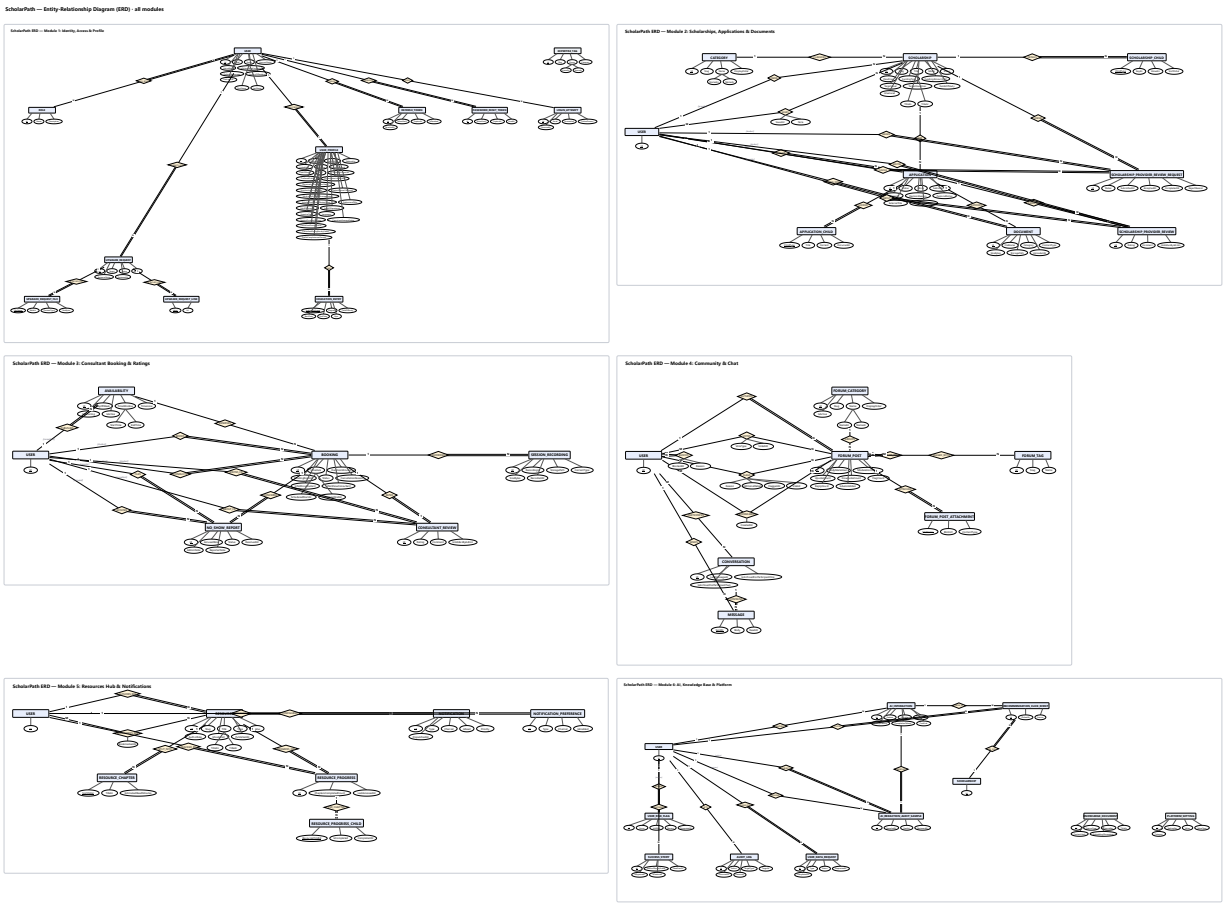


Figure 1: The basic conceptual ERD for ScholarPath (the "atlas" overview) in Chen / Elmasri notation — the deliberately plain starting point. `USER` is a single flat entity whose `Role` is just an attribute.

3.1 The core entities

The atlas gathers the platform's core entity types. Table 2 lists a representative selection with its meaning and the key attributes shown on the conceptual diagram.

Table 2: Representative core entity types of the basic ScholarPath ERD.

Entity	Meaning	Key / shown attributes
USER	Any account holder. At this stage Role is just an attribute distinguishing Student, ScholarshipProvider, Consultant and Admin.	Id, Email, Name, Role
CATEGORY	A subject classification applied to scholarships.	Id, Name
SCHOLARSHIP	A funding opportunity that students apply to.	Id, Title, Deadline, Status
APPLICATION	A student's submission against a scholarship.	Id, Status, SubmittedAt
DOCUMENT	A file uploaded to support an application.	Id, Category
SCHOLARSHIP_PROVIDER_REVIEW	A provider's rating of a reviewed application.	Id, Rating
AVAILABILITY	A time slot a consultant publishes for booking.	Id, IsActive
BOOKING	A booked consultant session.	Id, Scheduled-StartAt, Status
NO_SHOW_REPORT	A report that a booking party failed to attend.	Id, AccusedRole, Status
CONSULTANT_REVIEW	A student's rating of a completed booking.	Id, Rating
FORUM_POST	A post in the community forum.	Id, BodyMark-down, Modera-tionStatus
CONVERSATION	A private chat channel between two users.	Id
MESSAGE	A single message inside a conversation.	Id, Body
RESOURCE	A learning resource authored on the platform.	Id, Title, Type
NOTIFICATION	A message delivered to a user.	Id, Type, IsRead

3.2 The relationships and their cardinalities

Table 3 walks through the principal relationships drawn in Figure 1, giving the participating entities, the relationship verb and the cardinality ratio. Most of the backbone relationships are one-to-many; the many-to-many links (a user PARTICIPATES in many conversations, SAVES many scholarships, VOTES on many posts) are drawn as diamonds joining the two "many" sides.

Table 3: Relationships of the basic ScholarPath ERD with cardinality ratios.

From	Relationship	To	Ratio	Reads as
USER	OWNS	SCHOLARSHIP	1 : N	a provider owns many scholarships
CATEGORY	CLASSIFIES	SCHOLARSHIP	1 : N	a category classifies many scholarships
USER	SAVES	SCHOLARSHIP	M : N	a student saves many scholarships

From	Relationship	To	Ratio	Reads as
USER	SUBMITS	APPLICATION	1 : N	a student submits many applications
SCHOLARSHIP	FOR_SCH	APPLICATION	1 : N	a scholarship receives many applications
APPLICATION	SUPPORTS	DOCUMENT	1 : N	an application is supported by many docume
USER	OWNS_DOC	DOCUMENT	1 : N	a user owns many documents
APPLICATION	RATED_APP	SCH_PROV_REVIEW	1 : 1	an application is rated by one provider reviev
USER	PUBLISHES	AVAILABILITY	1 : N	a consultant publishes many availability slots
USER	BOOKS	BOOKING	1 : N	a student makes many bookings
USER	CONDUCTS	BOOKING	1 : N	a consultant conducts many bookings
AVAILABILITY	FOR_SLOT	BOOKING	1 : N	a slot reserves many bookings
BOOKING	CONCERNS	NO_SHOW_REPORT	1 : N	a booking may raise no-show reports
BOOKING	RATED_BKG	CONSULTANT_REVIEW	1 : 1	a booking is rated by one consultant review
USER	AUTHORS_POST	FORUM_POST	1 : N	a user authors many forum posts
USER	PARTICIPATES	CONVERSATION	M : N	users participate in many conversations
CONVERSATION	CONTAINS	MESSAGE	1 : N	a conversation contains many messages
USER	AUTHORS_RES	RESOURCE	1 : N	a user authors many resources
USER	RECEIVES	NOTIFICATION	1 : N	a user receives many notifications

A few observations are worth making explicit, because they shape the story of the later parts:

- **`USER` is the hub.** Almost every relationship has `USER` on one side. The same account plays the part of scholarship-provider owner (OWNS), student applicant (SUBMITS), consultant (PUBLISHES, CONDUCTS) and student booker (BOOKS). The diagram cannot tell these roles apart — they are all just "`USER`".
- **Two distinct roles on the same relationship target.** Both BOOKS and CONDUCTS connect `USER` to `BOOKING`; conceptually one end is a student and the other a consultant, but at this stage both are the same flat entity.
- **One-to-one review links.** RATED_APP and RATED_BKG are both 1:1, capturing that an application has at most one scholarship-provider review and a booking has at most one consultant review.

4 The Six Subject-Area Modules

The atlas is large, so the same plain ERD is also sliced into the six subject areas introduced in Section 1, one detail diagram per module. Each of Figures 2–7 zooms into a single module so the entities, attributes and cardinalities in that area can be read clearly. Every module keeps `USER` in view because almost every relationship touches it.

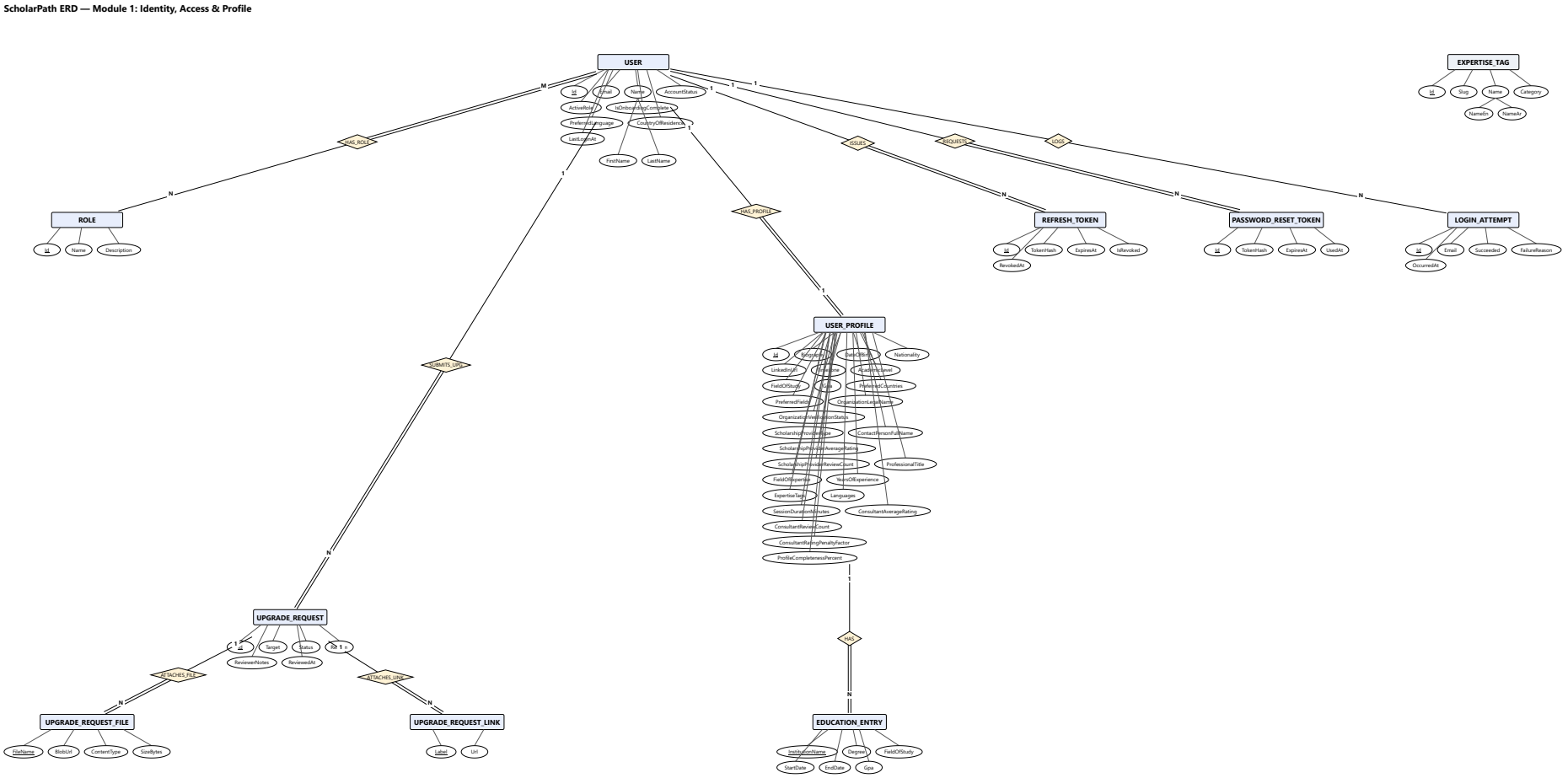


Figure 2: Module 1 — Identity, Access & Profile (plain ERD).

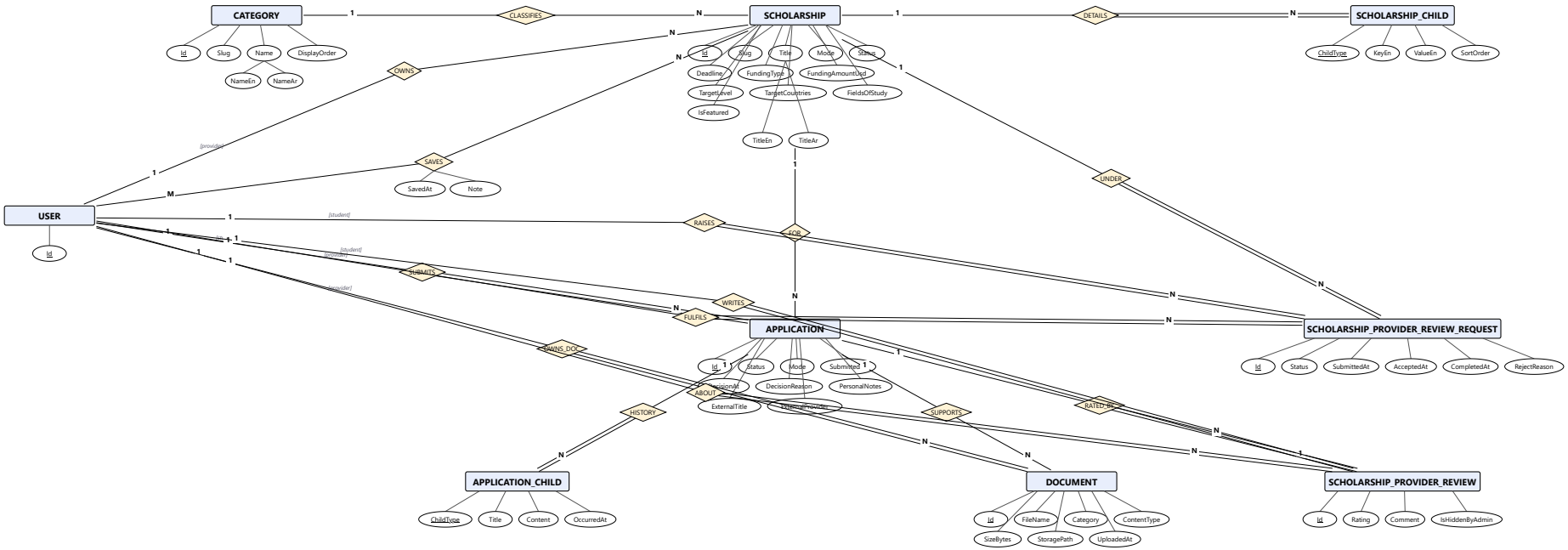


Figure 3: Module 2 — Scholarships, Applications & Documents (plain ERD).

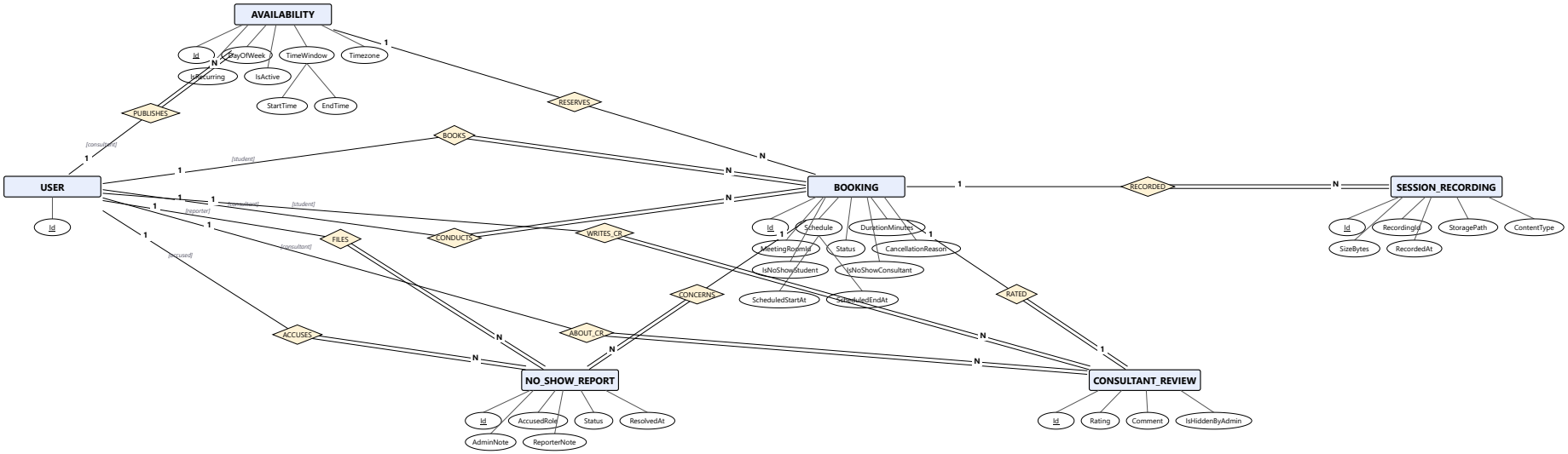


Figure 4: Module 3 — Consultant Booking & Ratings (plain ERD).

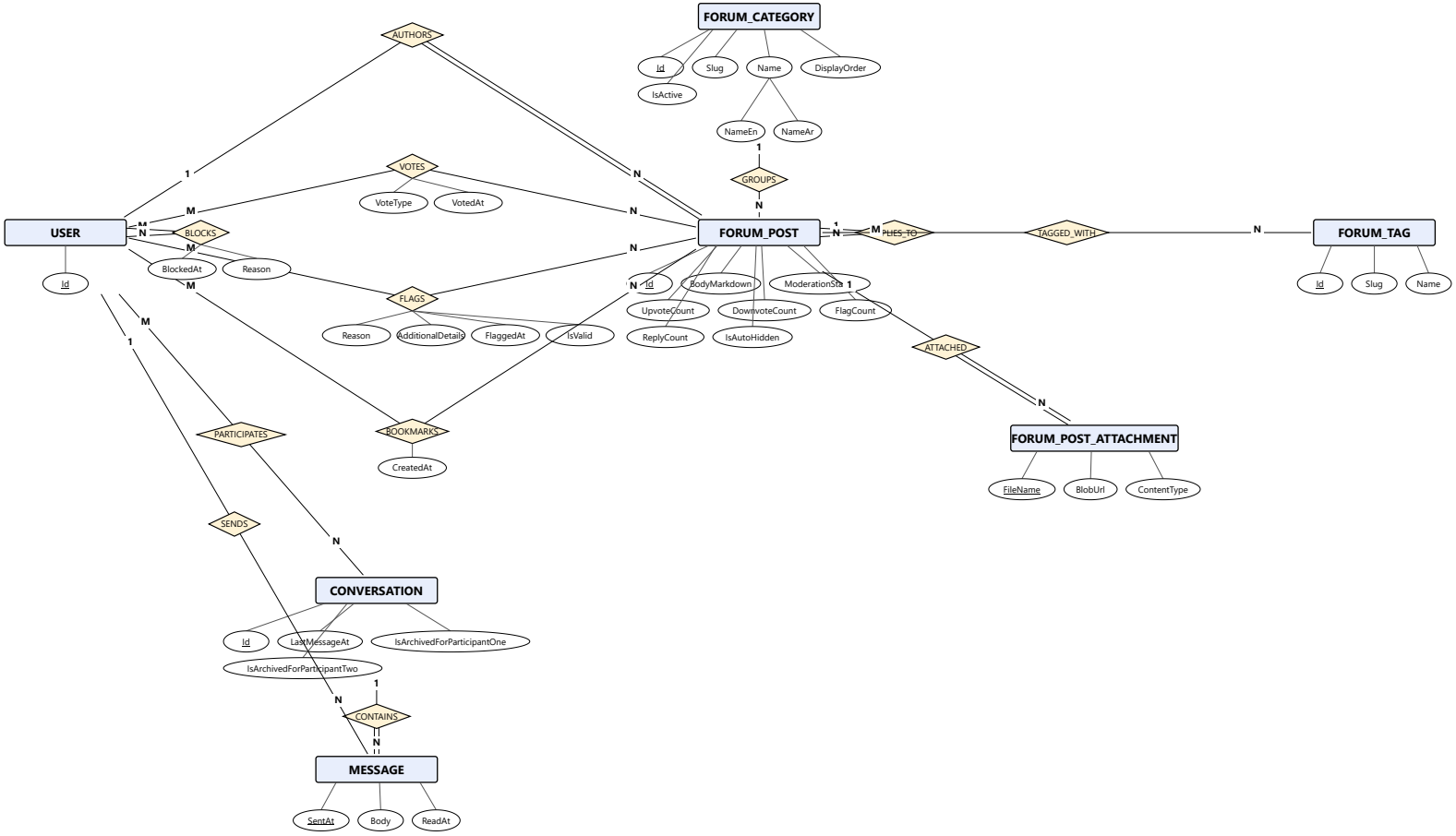


Figure 5: Module 4 — Community & Chat (plain ERD).

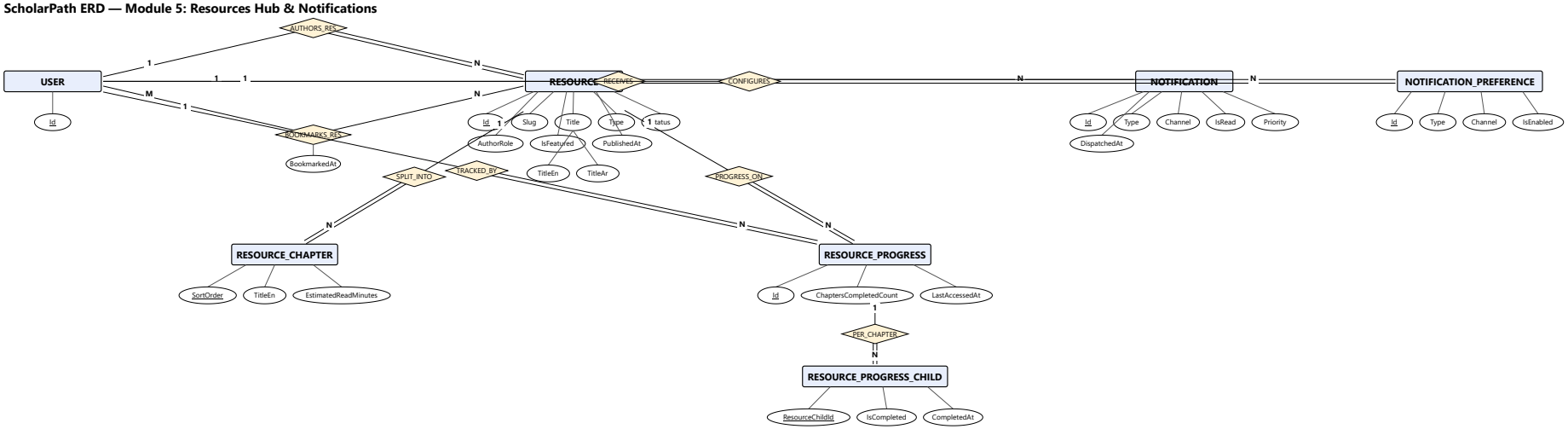


Figure 6: Module 5 — Resources Hub & Notifications (plain ERD).

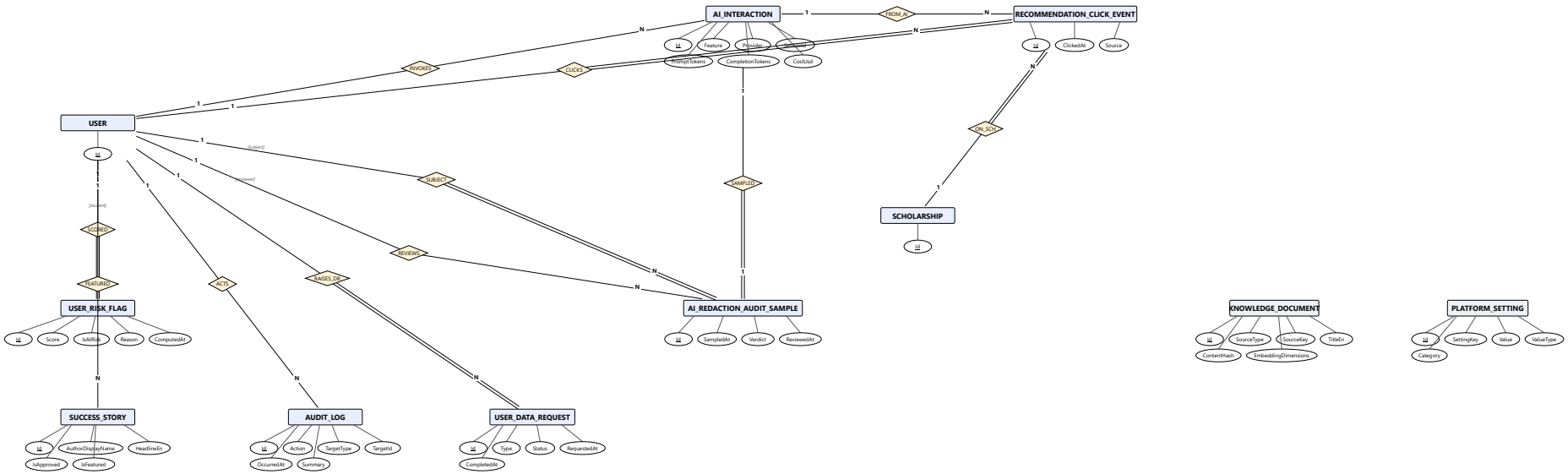


Figure 7: Module 6 — AI, Knowledge Base & Platform (plain ERD).

These per-module views are intentionally the same plain ERD as the atlas, only partitioned: they show *which* entities live in each area and how they relate, without yet committing to the enhancements. They are the map that the basic ERD of Figure 1 begins to fill in, and that Part 2 completes.

5 Limitations of the Plain ERD

The diagram in Figure 1 is deliberately the "before" picture. It is a faithful, readable first cut, but it leaves several real properties of the ScholarPath domain unexpressed. Naming these limitations is not a criticism of the model — it is precisely the agenda for *Part 2: The Enhanced ER Diagram*.

5.1 USER is one flat entity — role is just an attribute

The most important simplification is that there is a single `USER` entity, and the distinction between Student, ScholarshipProvider, Consultant and Admin is captured only by the `Role` attribute. This has visible consequences in the diagram:

- The role-specific attributes cannot be modelled. A student's academic level and GPA, a scholarship provider's organisation legal name and verification status, and a consultant's session duration and expertise all belong to *some* users but not others. A flat `USER` entity has no place to put them.
- The semantics of relationships are blurred. "A `USER` OWNS a `SCHOLARSHIP`" really means *a scholarship provider* owns it; "a `USER` BOOKS a `BOOKING`" really means *a student* books it. The diagram cannot enforce or even state these role constraints.

5.2 No specialization (no ISA hierarchy)

Because `USER` is flat, there is **no specialization / generalization**. The natural model is that `USER` is a super-type specialized into Student, ScholarshipProvider, Consultant and Admin sub-types, with the sub-type-specific attributes attached to each sub-type. The plain ERD has no way to draw the EER specialization circle or its disjoint/overlapping and total/partial constraints. Adding this ISA hierarchy is the headline enhancement of Part 2.

5.3 No weak entities

Every entity here is drawn as a strong (independent) entity with its own `Id` key. The model does not express that some entities exist only as *parts* of a parent — for example the per-chapter history of an application or the children of a scholarship. Such existence-dependent entities are properly modelled as **weak entities** (double rectangles, identified through their owner), a refinement introduced in Part 2.

5.4 No multivalued or derived attributes

All attributes here are simple single-valued ones. The domain, however, contains:

- **Multivalued attributes** — a consultant's expertise tags or a student's preferred countries are naturally a *set* of values, not a single value. Chen notation draws these as double ovals; the plain ERD omits them entirely.

- **Derived attributes** — a scholarship provider's average rating or a forum post's upvote count are *computed* from related entities rather than stored independently. Chen notation draws these as dashed ovals; again, the plain ERD does not distinguish them.

5.5 No distinction between enforced and loose references

Finally, the plain ERD treats every relationship as equivalent. In the real ScholarPath schema some associations are enforced by database foreign-key constraints while others are deliberately "loose" application-level references (used for high-volume audit, log and analytics data so that deleting a user never cascade-breaks history). That distinction is a modelling decision worth showing, and it too is introduced in the enhanced model.

5.6 Why this matters — the bridge to Part 2

Taken together, these limitations are the motivation for the next document. The plain ERD answers *"what are the things and how are they broadly related?"* The Enhanced ER Diagram answers the sharper questions the plain model cannot: *which* kind of user plays each role, which attributes are multivalued or derived, which entities are existence-dependent, and which references the database actually enforces.

Part 2: The Enhanced ER Diagram takes the entities of Figure 1 and enriches them: it specializes `USER` into its four sub-types, promotes the existence-dependent entities to weak entities, adds the multivalued and derived attributes, and annotates the enforced-versus- loose references. From there, *Part 3: Relational Mapping* turns the enhanced model into concrete tables, and *Part 4: Class Diagrams* expresses the same structure in object-oriented form. This document has set the stage by drawing the honest, deliberately plain starting point.